

LAB 6

DYNAMIC ROUTING CONFIGURATION: RIP, EIGRP, OSPF, BGP

OBJECTIVES

- To gain an understanding of how dynamic routing protocols such as RIP, EIGRP, OSPF, and BGP operate.
- To implement and configure these dynamic routing protocols on routers using Cisco Packet Tracer.
- To allow routers to automatically discover and update routes between multiple networks.
- To establish successful communication between networks through dynamic routing mechanisms

THEORY

1. RIP (Routing Information Protocol)

Operation:

RIP is a distance-vector routing protocol that determines the best path using hop count as its metric, with a maximum limit of 15 hops. It periodically sends the entire routing table to neighboring routers every 30 seconds and relies on the Bellman-Ford algorithm for route computation.

Application:

RIP is suitable for small-scale networks with simple structures. It is widely used in SOHO environments and academic labs due to its straightforward setup.

Limitations:

RIP supports only up to 15 hops, has slower convergence, consumes more bandwidth due to periodic updates, lacks VLSM support in RIPv1, and is prone to routing loops.

2. EIGRP (Enhanced Interior Gateway Routing Protocol)

Operation:

EIGRP is an advanced distance-vector protocol that uses the Diffusing Update Algorithm (DUAL) to calculate routes. Its metric is based on factors such as bandwidth, delay, reliability, and load. It sends incremental updates only when network changes occur and maintains neighbor relationships using hello messages.

Application:

EIGRP is well suited for large enterprise networks that require fast convergence, scalability, and hierarchical design. It can also be used in multi-vendor environments through route redistribution.

Limitations:

EIGRP is primarily Cisco-based, requires more processing resources, has more complex configuration than RIP, and troubleshooting can be challenging in large networks.

3. OSPF (Open Shortest Path First)

Operation:

OSPF is a link-state protocol that creates a complete map of the network topology. It uses Dijkstra's Shortest Path First algorithm to determine optimal routes. Networks are organized into areas with Area 0 serving as the backbone, and routers exchange Link-State Advertisements (LSAs) to maintain adjacency information.

Application:

OSPF is ideal for large and complex networks that demand high scalability and fast convergence. It supports hierarchical network design and works efficiently in environments with redundant paths.

Limitations:

OSPF involves more complex configuration, higher CPU and memory usage, and requires careful planning of areas. It also has a steeper learning curve for administrators.

4. BGP (Border Gateway Protocol)

Operation:

BGP is a path-vector protocol used to exchange routing information between autonomous systems. It operates over TCP port 179 and determines routes based on policies, rules, and path attributes rather than simple metrics.

Application:

BGP is essential for Internet routing and is commonly used to connect organizations to ISPs, support multi-homing, and implement policy-based routing between different networks.

Limitations:

BGP configuration is highly complex, convergence is slower compared to IGP, it requires significant memory for large routing tables, and misconfigurations can have widespread network impacts.

NETWORK TOPOLOGY

The designed network topology includes two routers—Router0 (ISR4331) and Router1 Sachyam (ISR4331)—connected through a serial WAN link using their serial interfaces. This setup forms two separate Local Area Networks (LANs) that communicate through dynamic routing.

On the left side, Switch0 (2960-24TT) is connected to Router0 Sachyam and provides network access to two end devices, PC0 and PC1, creating the **192.168.1.0/24** LAN.

On the right side, Switch1 (2960-24TT) is connected to Router1 Sachyam and connects PC2 and PC3, forming the **192.168.2.0/24** LAN.

The serial connection between the two routers enables communication between these networks, allowing devices in both LANs to exchange data through dynamic routing protocols. In the Cisco Packet Tracer simulation, the green link indicators confirm that all connections are active and functioning properly.

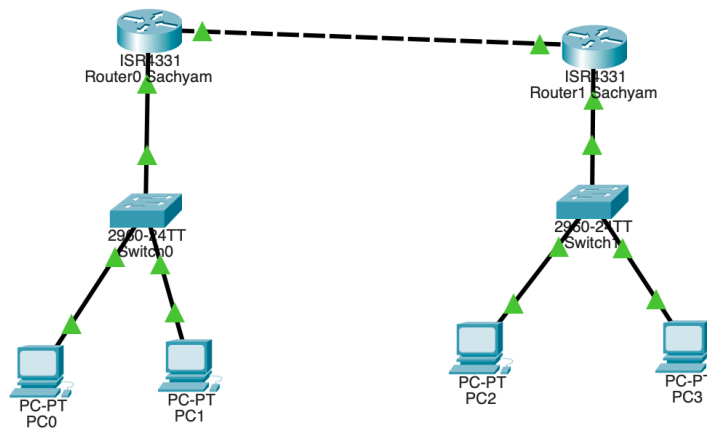


Fig: Network Topology

Configuration Table:

Network devices were configured with IP addresses and subnet masks as detailed in the table below:

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0 Sachyam	Gig0/0/1	192.168.1.1	255.255.255.0	N/A
Router0 Sachyam	Gig0/0/0	10.0.0.1	255.0.0.0	N/A
Router1 Sachyam	Gig0/0/1	192.168.2.1	255.255.255.0	N/A
Router1 Sachyam	Gig0/0/0	10.0.0.2	255.0.0.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

1. RIP Configuration Commands

Router0 Sachyam (Left side in the topology)

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Router1 Sachyam (Right side in the topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

2. EIGRP Configuration Commands

Router0 Sachyam (Left side in the topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Router1 Sachyam (Right side in the topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.2.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

3. OSPF Configuration Commands

Router0 Sachyam (Left side in the topology):

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Router1 Sachyam (Right side in the topology):

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

4. BGP Configuration Commands

Router0 Sachyam (Left side in the topology) — AS 65001:

```
Router(config)# router bgp 65001
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
Router(config-router)# network 192.168.1.0 mask 255.255.255.0
```

Router1 Sachyam (Right side in the topology) — AS 65002:

```
Router(config)# router bgp 65002
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

RESULT

The dynamic routing protocols were configured and tested successfully within the designed network topology. Connectivity was verified by sending ping requests from PC2 (192.168.2.2) in the 192.168.2.0/24 network to PC0 (192.168.1.2) in the 192.168.1.0/24 network.

Each protocol—RIP, EIGRP, OSPF, and BGP—demonstrated proper automatic route exchange and updates to the routing tables. The successful transmission and reception of packets without loss confirmed effective inter-network communication and validated that all dynamic routing configurations were correctly implemented

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: ping (RIP)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: ping (EIGRP)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: ping (OSPF)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: ping (BGP)

DISCUSSION

This experiment effectively demonstrated the configuration and operation of four dynamic routing protocols using Cisco Packet Tracer. RIP offered the simplest setup but showed slower convergence due to its periodic updates. EIGRP provided faster convergence and better bandwidth efficiency through its triggered update mechanism. OSPF required more detailed configuration, including area planning, but delivered a standardized and scalable routing solution. BGP involved the most complex configuration because of its autonomous system structure and policy-based routing approach.

CONCLUSION

The laboratory exercise successfully achieved its goal of implementing and understanding dynamic routing protocols on Cisco routers. The experiment highlighted how dynamic routing enables automatic route discovery and maintenance, eliminating the need for manual route configuration while ensuring efficient and reliable communication across networks.

